

M.C.A. Integrated (Semester-VIII) Examination - June - 2026
IMCA-801T
Machine Learning

Time Allowed: Three Hours

Max. Marks: 70

This question paper contains two sections as under :

Min. Marks: 28

Section-A

Max. Marks-20

This section will have 10 questions, having 02 questions from each unit. Answer in about 20 words. All questions will have equal marks.

(2×10=20)

Section-B

Max. Marks-50

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400 words. Questions may have sub divisions. All questions will have equal marks.

(10×5=50)

Section-A

1. (i) Define overfitting.
- (ii) What is machine learning?
- (iii) What is random variable?
- (iv) What is probability?
- (v) What is regression?
- (vi) What is classification?
- (vii) What is supervised learning?
- (viii) Write main limitations of decision tree.
- (ix) What is clustering?
- (x) What is validation?

Section-B

Unit-I

2. Explain :
 - a) K-nearest neighbour method
 - b) Validation
 - c) Density estimation

OR

Discuss different types of machine learning algorithms with their properties.

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Unit- II

3. Explain Bayes' theorem with a suitable example.

OR

✓ Explain linear classifier with an example

Unit - III

4. ✓ Explain importance of Support Vector Machines (SVMs) and also discuss functioning of SVMs.

OR

Discuss Logistic regression with an example. - *clarify*

Unit- IV

5. ✓ Discuss Key advantages and disadvantages of using decision tree and its functioning.

OR

Discuss mathematical criteria adopted in decision tree classifiers and also discuss about its consistency.

Unit- V

6. Explain K. means clustering algorithm with its limitations.

OR

✓ Explain agglomerative algorithm with a suitable example.
